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1. What are some considerations when optimising the composition of F/M Steels? (10 Pts)

A: When adding or subtracting concentrations of Alloying elements, some of the F/M Steel's Performance will be affected. An example is the Tuning of Cr in the Alloy System, too much Cr leads to undesirable phases, too little Cr can lead to degradation of High T Corrosion Performance, considering the Formation of phases and the resulting Material Performance is key when optimising F/M Steels.

- needed a bit more consideration

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2. Why do ferritic Steels swell less than austenitic Steels? (10 Pts)

A: Ferritic Steels have an BCC Phase ~~than~~ which allows for more ^{high reduction volume} space for interstitials & Vacancies, this is due to the Atomic Packing Factor of BCC, which is lower than FCC. Additionally the Migration energy is less than Austenitic Steels, which allows for ^{more} recombination in ferritic Steels, mostly the formation of carbides in ferritic Steels, allows for sinks further increasing the resistance to swelling.

10/10

3. What is the Ductile to Brittle Transition Temperature? Discuss irradiation embrittlement in F/M alloys? (10 Pts)

A: The Ductile to brittle Transition Temperature ^{should be more explicit} is the Temperature range where a material's Mechanical behavior changes. Commonly, DBTT is measured by Charpy Energy (Y-axis) compared to Temp (X-axis), when considering Irradiation, DBTT is affected. because ~~large~~ Irr. damage causes an increase in DBTT. Radiation embrittlement in F/M Steels changes the Phases present in the ~~alloy~~ alloy to more brittle Phases. The accumulation of defects can also decrease the ductility of the Steel due to the increase in dislocation density.

- some of it ..

7/10

4. Why are FeCrAl alloys of interest? What are the considerations for balancing compositions of these alloys? (8 pts)

A: FeCrAl alloys are an ~~interest~~ ^{oxidation!} interest because of their irradiation performance and lack of Zr. This allows for an ideal cladding system that can replace Zircaloy to prevent the failure that happened in Fukushima. When balancing the composition of FeCrAl alloys it is important to consider both the corrosion and mechanical performance. Adding too much Al or Cr can cause undesirable phases to form, which can negatively effect mech. performance. Less Al or Cr can prevent the formation of the oxidation layer which protects the alloy from further corrosion. 7/8

5. What role do Oxide Particles Play in ODS steels? (6 pts)

A: The Oxide Particles act as a strengthening mechanism in the material. The Oxide Particles hinder the movement of dislocations & can act as sinks for vacancies & interstitials. These mechanisms allow for a High T resistance to creep & irradiation damage, while stabilizing the microstructure. 6/6

6. What are some advantages & disadvantages of Ni alloys? (2 pts)

A: Ni based alloys can be used at higher T's due to their ^{phase} stability. 8/8
Furthermore, Ni based alloys are more compatible with molten salt & liquid metals because of their inert composition. (little Cr leads to less Cr depletion)

However, the transmutation of Zr leads to alpha (He) particles. This leads to He gas being produced which is not soluble in the alloy. The He will eventually lead to embrittlement of the material w/ increased flux & T.

7. What are the benefits & drawbacks of U-Si fuel compared to U-Al fuel? Describe the differences in U3Si vs U3Si2 swelling. (12 pts)

Benefits
Higher fissile density
Better swelling performance
IL can accommodate gas products

- U-Al shows negligible swelling

drawbacks
• Difficult to manufacture.
• Undesirable behavior can form when different phases form.

• Amorphization occurs in all phases at low T inn. - for all U-Al & U-Si

U₃Si vs U₃Si₂
more accommodation of swelling due to stoichiometry
- needed more here...

8. The gamma phase of U-Mo is not thermodynamically stable phase at PSC reactor temperatures. Why is this phase ~~stable~~ the dominant phase in-reactor? (7 pts)

A! The reason this phase is dominant in reactor conditions even though it's not thermodynamically stable is due to radiation induced phase ~~segregation~~ defect generation causes the conditions that allow the gamma phase to form. ✓
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9. What effect does the solidus/liquidus gap have ~~on the formation~~ on the fabrication of U-Mo fuels? (6 pts) ✓

A! The effect the solidus/liquidus gap has on the fabrication of U-Mo fuels is the ~~the~~ homogeneity of the microstructure. Mo enriched zones can decrease the fuel's fission density and are undesirable phases. - not different phases

10. Discuss the evolution of fission gas bubbles in U-Mo fuel as a function of burn-up. (10 pts)

A: low burn up: Fission gas bubble form a sub-lattice ✓ & bubble accumulate @ grain boundaries (GB) - not really...
7/10

Mid burn up: Fission gas bubbles form more @ GB & Grain coarsening begins? - grain refinement, opposite of coarsening

High burn up: grains fully coarsen, Fission gas bubbles dominate across ~~the~~ fuel microstructure. ✓

11. Why is the Interfacial layer of interest in U-Mo fuels? What mitigation strategies have been deployed to limit the amount of the interfacial layer? (7 pts)

The IL degrades the fuel performance, as phases with less fission density form. Furthermore other properties such as thermal conductivity are negatively affected. The main & most effective mitigation strategy is to use a monolithic fuel design that eliminates the need for an Al matrix. Instead a Zr foil is used. The fuel is more fission dense as well, - not necessarily

- IL amorphization & swelling

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1/2. Why is Al ideally suited for research reactor environments when it is unable to be used in LWRs? (6pts)

A: The operating T of RR is about $0.3 T_m$ of Al. This allows for ideal irradiation performance at lower T 's. Al is a good thermal conductor, corrosion resistant & is cheap/easy to manufacture.

✓